

PUBLISHED

Question	Answer	Marks	Guidance						
5	H_0 : performance is independent of age. H_1 : performance is not independent of age. OR H_0 : passing test is independent of age. H_1 : passing test is not independent of age.	B1	Allow dependent, association or relationship for not independent. For example: H_0 : No association/relationship between... H_1 : Association/relationship between... Must refer to “performance” and “age” in H_0 .						
	<table border="1"><tr><td>34 (27.93)</td><td>41 (44.69)</td><td>6 (8.38)</td></tr><tr><td>16 (22.07)</td><td>39 (35.31)</td><td>9 (6.62)</td></tr></table>	34 (27.93)	41 (44.69)	6 (8.38)	16 (22.07)	39 (35.31)	9 (6.62)	M1	At least 3 correct to 3sf (must be seen).
	34 (27.93)	41 (44.69)	6 (8.38)						
	16 (22.07)	39 (35.31)	9 (6.62)						
		A1	All correct to 3sf.						
	$\frac{(34 - 27.93)^2}{27.93} + \frac{(41 - 44.69)^2}{44.69} + \dots + \frac{(9 - 6.62)^2}{6.62}$ $[1.319 + 0.305 + 0.676 + 1.669 + 0.386 + 0.856]$ OR $\left(\frac{34^2}{27.93} + \frac{41^2}{44.69} + \dots + \frac{9^2}{6.62} \right) - 145$	M1	At least 2 expressions (not including 145) of correct form seen. May see individual terms not summed. May be implied by 2 contributions correct to 3sf seen. May be implied by AWRT 5.21.						
	5.21	A1	AWRT 5.21.						
'5.21' > 4.605 , reject H_0 / significant.	M1	Compare <i>their</i> 5.21 with 4.605 and appropriate result (may be in terms of H_1). Test result may be implied by an attempt at an appropriate conclusion in context ignoring hypotheses.							
Sufficient evidence to suggest that performance and age are not independent. OR Sufficient evidence to suggest that performance and age are dependent. OR Sufficient evidence to suggest that performance and age are associated. OR Sufficient evidence to support the (driving) instructor’s belief.	A1	Correct conclusion from correct working ignoring hypotheses. Must refer to both performance and age. In context. Level of uncertainty in language is used (for example, not ‘prove’). Do not accept statements such as ‘there is insufficient evidence to suggest...’.							
	7								